Segmentation & Representation Benchmark Matrix

Consolidated validation across three corpora (MeerKAT, LibriSpeech, InfantMarmosetsVox), empirical

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Terminology — what the pipeline discovers in animal audio

Speech word-discovery methods (DUSTED, ES-KMeans, Bayesian word segmentation, SSL-unit clustering

speech (human, has GT)	animal (no GT meaning)
word	recurrent acoustic term / motif
word-like unit	motif-like acoustic term
phoneme / subword	sub-call acoustic unit
lexicon	motif inventory
sentence	bout / vocalisation sequence

All calltype/caller "enrichment" reported below is **acoustic/structural association, not semantics

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1. Event-detection F1 (which segmenter, which regime)

corpus (regime)	method	metric	F1	P	R	ref
MeerKAT (callsign noise, 8 kHz)	raw NBD	onset 1-to-1	0.069	—	—	massive over-segmentation
	sweep-best threshold	onset 1-to-1	0.165	—	—	thresholds insufficient
	v2 scalar GBT (19 feat)	onset 1-to-1	~0.26	—	—	tabular baseline (frozen)
	v3/v4 post-onset GBT	onset 1-to-1	~0.21	—	—	tabular **ceiling**
	PCEN+CRNN	onset 1-to-1	**0.703**	—	—	**8.9× raw NBD → gap closed**
LibriSpeech (scalar GBT speech, word-level)	scalar GBT speech, word-level	word 1-to-1 / combined	0.450 / 0.638	—	—	CPC-free
	CPC-context GBT (+dim)	word 1-to-1 / combined	0.458 / 0.794	—	—	CPC adds +0.156 combined
	NBD+CPC calibrated	**combined (§1 metric)**	**0.836**	0.77	0.92	**speech winner**
	PCEN+CRNN (retrained)	word 1-to-1 / combined	0.590 / 0.652	0.80	0.47	CRNN is the **speech bottleneck**
InfantMarmosetsVox (clean, adult)	**PCEN+CRNN (retrained)**	onset 1-to-1	**0.875**	**0.982**	0.789	**strongest result**

Router read: PCEN+CRNN wins for discrete calls in noise (0.70 meerkat → 0.88 marmoset); NBD+CPC

Tolerance sweep (apples-to-apples — `scripts/tolerance_sweep.py`)

The headline F1s used different tolerances (MeerKAT 100ms, marmoset 200ms). At **matched**

tolerance

onset F1	50ms	100ms	200ms
MeerKAT (500 clips)	0.662	0.707	0.724
InfantMarmosetsVox (30 test)	**0.864**	**0.865**	**0.867**

offset F1	50ms	100ms	200ms
MeerKAT	0.549	0.664	0.702
InfantMarmosetsVox	**0.785**	**0.860**	**0.866**

Marmoset onset F1 is tolerance-*insensitive* (0.864→0.867) because precision ≈ 0.98 (near-

zero false

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2. HECB internal-boundary mechanism (oracle-parent, no CRNN)

The internal mechanism decoupled from the detector — given valid parent events, does HECB

recover m

corpus	parent events	sub-event recovered	F1	P	R
LibriSpeech	oracle word spans	in-word phoneme start	**0.636**	0.781	0.537
LibriSpeech	CRNN word events	in-word phoneme start	0.409	0.516	0.339

Read: the mechanism works at 0.636 with good parents but degrades to 0.41 with poor

(CRNN) pa

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3. Whole / sub-call / pooled representation recovery

The core claim: sub-call features **augment** whole-event representations when pooled, but

are weake

Call-type recovery

corpus	split	whole V	pooled V	whole probe	pooled probe	whole macroF1	pooled macroF1
MeerKAT (end-to-end, 1500 clips)	—	0.246	**0.274**	0.611	**0.633**	—	—
InfantMarmosetsVox (end-to-end)	—	0.344	**0.379**	0.934	**0.940**	—	—
InfantMarmosetsVox (oracle)	random	0.431	0.416	0.935	0.932	0.487	**0.566**
InfantMarmosetsVox (oracle)	within-callid	0.477	0.475	0.944	**0.947**	0.557	**0.582**

*Sub-call alone is consistently weaker on V (loses whole-call context) but highest on probe

(sub-fragm

Caller-ID recovery (InfantMarmosetsVox oracle, random split, 10 callers) — **the novel finding**

representation	V	NMI	probe	macro-F1
whole call	0.178	0.070	0.459	0.364
whole + sub-call pool	**0.197**	**0.083**	**0.477**	**0.392**
sub-call	0.115	0.029	0.317	—

HECB internal boundaries recover caller identity — pooled wins on all four metrics.

Individual

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4. The pooled-adds-value thesis (cross-corpus summary)

corpus	label	pooled > whole?	on which metric
MeerKAT	call-type		V, NMI, probe (clean)
InfantMarmosetsVox (end-to-end)	call-type		V, NMI, probe
InfantMarmosetsVox (oracle)	call-type	mixed	macro-F1 (+0.079); whole edges V
InfantMarmosetsVox (oracle)	**caller ID**		**all 4 metrics**

Refined conclusion (post-controls): Sub-call structure augments whole-call

represent

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5. Segmentation router (canonical)

regime	method	HECB role	example
discrete calls in noise	PCEN+CRNN	internal annotator	meerkat (0.70), marmoset (0.88)
continuous speech-like	NBD+CPC calibrator	optional	LibriSpeech (0.836 combined)
exploratory / unknown	NBD + ABSTAIN	none	new species (conservative)

Updated core claim (project thesis): The pipeline uses a regime-specific segmentation

router: PC

Caveats

All F1s are one-to-one greedy matching (no duplicate-TP inflation). MeerKAT onset

toleranc

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6. Cross-species 4-way ablation (raw-NBD oracle, 2026-06-22)

`scripts/cross_species_ablation.py` — per-file oracle-parent HECB (raw NBD internal

boundarie

corpus (kHz)	label	whole V	pooled V	**sequence V**	whole probe	pooled probe	seq probe
**adult marmoset (vocalization, 96)	call-type (6)	0.458	0.434	**0.468**	0.824	0.832	**0.837**
**egyptian fruit bat (250)	Context (2)	0.152	0.152	0.153	0.547	0.556	0.543
egyptian fruit bat (250)	Emitter (ID)	0.291	0.291	0.291	0.211	0.191	0.195

Honest read — this REFINES the pooled-adds-value thesis:

**The re

Revised cross-species claim: sub-call structure adds value where (a) the internal

boundarie

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7. Typed HECB vs raw NBD on adult marmoset (boundary-quality test, 2026-06-22)

`scripts/typed_hecb_vs_raw.py` — same eval files (2491 calls, 6 types), two internal-

boundary
lacked.

method	whole V	pooled V	pooled−whole	pooled probe	sub-call V	sub-call probe
raw NBD	0.450	0.418	−0.032	0.800	0.173	0.487
typed HECB	0.450	**0.431**	**−0.019**	**0.814**	**0.255**	**0.527**

Honest result: typed HECB improves sub-call fragment quality (V 0.255 vs 0.173, probe

0.527 vs (

Cross-experiment pattern (now consistent): the sub-call advantage appears on

**supervis

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8. Caller-ID confound controls (InfantMarmosetsVox, 2026-06-22)

`scripts/caller_id_controls.py` — validates the novel caller-ID finding against confounds.

2828 vocs

Baseline + within-call-type (V / NMI / probe)

condition	whole V	pooled V	pooled−whole V	whole probe	pooled probe
all call types	0.185	**0.203**	+0.018	0.452	**0.479**
within Phee (10 callers)	0.223	**0.234**	+0.011	**0.529**	0.465

within Twitter (10 callers)	0.251	**0.261**	+0.010	0.594	**0.601**

Trivial baselines (probe accuracy)

baseline	probe acc	macro-F1	interpretation
duration-only	0.165	0.070	near-chance (0.10) — **NOT duration**
calltype-onehot	0.138	0.059	near-chance — **NOT call-type confound**
whole-fp (dur-regressed)	0.435	0.393	survives duration removal

Verdict: the caller-ID finding is **genuine** — it survives all confound controls.

The poole

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9. Track C bat context ablation (acoustic + social + behavioral, 2026-06-22)

`scripts/bat_trackc_ablation.py` — the scientific gap: does combining modalities recover

context b

Context recovery — the A–G pairwise ablation matrix (the key target)

model	features	probe acc	macro-F1	lift
A	acoustic-only (fingerprint)	0.249	0.218	2.7×
B	social-only (emitter+addressee)	0.267	0.223	2.9×
C	behavior-only (pre/post actions)	0.087	0.162	1.0×
D	acoustic + social	**0.393**	**0.341**	**4.3×**
E	acoustic + behavior	0.298	0.307	3.3×
F	social + behavior	0.369	0.327	4.1×
G	acoustic + social + behavior	**0.424**	**0.355**	**4.7×**

The pairwise decomposition reveals where the synergy lives:

**D (aco

Leakage / generalization controls (reviewer-proof)

split	A acoustic	B social	D acous+social	G full
random	2.7×	2.9×	**4.3×**	**4.7×**
social-shuffle (emit/addr permuted)	2.7×	**1.0×**	**3.2×**	4.0×

emitter-held-out (GroupShuffleSplit)	2.6×	2.8×	**4.1×**	**4.8×**

Social-shuffle collapses B to chance (1.0×) and drops D by 1.1× (4.3→3.2) → ~69% of D's lift o

Manuscript-ready Track C conclusion

Acoustics encode call form and caller-associated structure, but ethological context requires *

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10. Sub-call motif structure report (InfantMarmosetsVox, 2026-06-22)

`scripts/subcall_motifs.py` + `scripts/motif_controls.py` — 2423 vocalisations, k=12 fingerprin

Position-specialized clusters (onset/body/offset roles)

cluster	first%	middle%	last%	**role**	calltype V	caller V
7	**84.0**	15.7	0.2	**onset**	—	—
1	0.6	4.2	**92.4**	**offset**	—	—
0	2.3	**97.6**	0.1	**body**	—	—
10	0.0	**100.0**	0.0	**body**	—	—
11	0.2	**99.8**	0.0	**body**	—	—
3	29.7	**68.3**	2.0	body (onset-leaning)	—	—

Calltype-enriched motifs (top by V)

level	motif	count	calltype V	p
1-gram	cluster 0	1408	0.653	<0.0001
1-gram	cluster 7	1092	0.650	<0.0001
2-gram	(0, 0)	462	0.398	<0.0001
2-gram	(0, 7)	307	0.372	<0.0001
3-gram	(11, 11, 11)	175	0.242	<0.0001
3-gram	(0, 7, 0)	104	0.227	<0.0001

Control summary

control	result
bootstrap stability (bigram Jaccard)	**1.000** (STABLE)
bootstrap stability (trigram Jaccard)	**0.773** (STABLE)
position enrichment	onset/body/offset roles confirmed (84%/9
within-Phee caller motifs	(10,0) V=0.186 p<0.0001 — caller signal
bigram-preserving trigram null	374 vs 381±5 (5th pct) — trigrams ≈ bigr
token-shuffle bigram richness	94 vs 104±1 — **non-random** (constraine

Manuscript-ready claim

HECB-derived sub-call units are stable, non-random, and positionally specialized.

Cluster 7

11. DUSTED-style recurrent-subsequence motif discovery (3-way method comparison, 2026-06-22)

Adapts spoken-term-discovery (DUSTED) to **recurrent acoustic terms** in animal audio

(`diagnos

exact n-grams — windows match iff cluster-ID tuples are identical (fragments every

substitut

Run: `python scripts/hecb_dusted_marmoset.py --n-eval 1500` (`scripts/hecb_dusted_marmoset.py`).

Method comparison (the decisive question — variant tier, DTW thr=0.35)

method	#motifs	total support	coverage	mean calltype V	mean caller V
exact n-grams	376	6236	0.155	0.109	0.102
fixed-length cosine	18	5988	0.509	0.289	0.125
DTW with indels	**2**	**6970**	**0.516**	**0.508**	**0.170**

DTW merges variants into the fewest, richest families: highest support/coverage and

**calltype

Honest caveats (do not overclaim)

1. **Degenerate inventory.** InfantMarmosetsVox sub-call structure is dominated by ONE

recurren

2. **Order-shuffle null is conservative for repeats.** The within-call order shuffle

preserv

3. **Bootstrap Jaccard low (0.16–0.18).** With a degenerate inventory (1 dominant + sparse minors)

Manuscript-ready claim

DTW-with-indels is the right subsequence-matching primitive for recurrent-acoustic-term discovery

12. Bengalese-finch song — the rich-repertory positive control (GT-validated, 2026-06-22)

The marmoset caveat (§11) predicted a richer inventory from a richer corpus. Bengalese finch song

3-way method comparison (mean across Bird0/1/2)

method	#motifs	coverage	label purity (mean / wmean)
exact n-grams	106	0.84	1.00 / 1.00
fixed-length cosine	104	0.90	0.97 / **0.895**
DTW with indels	81	0.90	0.83 / 0.59

The expected, clean method trend — and crucially, **fixed-length cosine recovers the GT syllable m

DTW recovers real syllable motifs blind from acoustics (selected, per bird)

bird	motif (GT labels)	support	purity	note
Bird1	`[1,0]`	304	**0.90**	clean 2-syllable motif across 141 cal
Bird2	`[4,5]`	158	**1.00**	perfectly recovered bigram, 118 calls
Bird2	`[5,6]`	148	0.78	—
Bird2	`[10,11,12,13]`	209	0.51	4-syllable motif
Bird0	`[2,3,4]` / `[6,2,3]` / `[5,6,15]`	6–10	0.51–0.56	trigrams

Nulls + stability (per bird)

Observed DTW support is **2.0–2.3× the order-shuffle null** on every bird (p=0.111 is the

n_perm=8

Manuscript-ready claim

On Bengalese-finch song — the right dataset class for diverse recurrent acoustic terms —

DUSTED c

13. Egyptian fruit bat — emitter/social context; graded, not discrete (2026-06-22)

The stress test the Bengalese result (§12) set up: does a *socially annotated* corpus

form recu

The decisive finding: Emitter V ≫ Context V (every method)

method	#fams	coverage	H(norm)	top-frac	**Emitter V**	**Context V**
exact n-grams	582	0.29	0.91	0.04	0.463	0.164
fixed-length cosine	155	0.78	0.60	0.43	0.454	0.167
DTW with indels	105	0.82	0.32	0.72	0.467	0.163

**Emitter V ≈ 0.45–0.47 vs Context V ≈ 0.16–0.17 (~2.8×) — rock-solid across all three

methods.*

Bat is graded/intermediate, not discrete-syllable like Bengalese

The diverse-vs-degenerate diagnostic (the user's `family_entropy` / `top_family_support_fraction`):

bat sits **

Honest caveats

Null modest (obs/null ≈ 1.23×, p=0.2 at n_perm=4): the dominant `[12,12]` repeat is

composi

Manuscript-ready claim

On Egyptian fruit bat, recurrent acoustic-term families are tied to **emitter/caller

(0.45–0.4

14. Unsupervised word-structure benchmark — HECB vs SSL discrete units (LibriSpeech, 2026-06-22)

The decisive external comparison: is HECB's boundary-aware internal sub-unit sequence

competiti

Headline table

unit source	within	across	a_dur	a_phn	ratio	NN acc	NN_xspk	motif→word V
HECB	0.908	0.930	0.939	0.929	0.976	0.089	0.085	0.326
HuBERT-km(50)	0.928	0.920	0.924	0.923	**1.008**	0.052	0.052	0.302
CPC-km(50)	0.900	0.943	0.944	0.943	**0.954**	**0.191**	0.178	0.322
fingerprint(328)	0.897	0.931	0.934	0.933	0.964	0.147	0.142	**0.347**

Controls (all clean)

`a_dur` / `a_phn` (across-word baseline matched on duration / phoneme-count) ≈ `across`

for every

Findings

1. **CPC-km wins** raw same-word consistency (ratio 0.954, NN 0.191). The pipeline's own

SSL mo

Honest caveats (do not overclaim)

HuBERT's failure is metric-specific, not a defeat of HuBERT as a representation.

The zero

Manuscript-ready claim

On a controlled LibriSpeech same-word-consistency benchmark, HECB's boundary-aware

sub-unit s

> **Fingerprint note (all experiments):** the `fingerprint` source is `afp.extract(...).combined`

> = the
species-n

15. Stage C + D — DUSTED term discovery & ES-KMeans clustering (HuBERT under DTW; LibriSpeech, 2026-06-23)

Two follow-on stages closing the §14 "HuBERT fails" loop (was it the metric, not the unit?).

Both reus

Stage C — DUSTED word-term discovery (DTW, HuBERT's native metric; `dusted_librispeech_words.py`)

Each word instance = one CallUnits (discrete-unit sequence, `calltype=word`). DUSTED module run

per source

unit source	exact word_V	DTW word_V	DTW fams	DTW cov	top-frac

HECB(k32)	0.409	0.425	254	0.048	0.02
HuBERT-km(50)	0.415	0.418	501	0.122	0.02
CPC-km(50)	0.418	0.431	651	0.075	0.01
fingerprint(32)	**0.448**	**0.458**	259	0.078	0.03

HuBERT recovers under DTW: word_V 0.418 (vs §14's Levenshtein *failure* ratio 1.008). So

§14's "Hu

Stage D — whole-word embedding clustering (ES-KMeans lexicon core; `eskmeans_librispeech.py`)

Oracle GT word spans → whole-word acoustic embedding → k-means (k=60) → evaluate vs GT word

labels. Co

embedding	V-measure	ARI	purity	token-F1
fingerprint (44-D)	**0.481**	0.061	**0.276**	**0.276**
CPC-mean (256-D)	0.436	0.036	0.242	0.242
HECB-pooled (44-D)	0.392	0.012	0.198	0.198
HuBERT-mean (768-D)	0.338	−0.003	0.172	0.172

Fingerprint clusters words best (V=0.481, token-F1 0.276 = 1.6× chance over 88 types);

HuBERT-m

Manuscript-ready claim

Across four complementary LibriSpeech word-structure tests (§14 Levenshtein, Stage C DTW term

discovery

16. HuBERT sanity ablation — the conclusion is locked (2026-06-23)

The reviewer-safe sweep: does ANY HuBERT configuration (layer × k × collapse × pooling)

beat the f

Track 1 — sequence consistency (18 configs: layer ∈ {6,9,12} × k ∈ {50,100,200} × {col,raw})

Every single HuBERT config has ratio **1.001–1.018** (within ≥ across: *no* same-word structure)

Track 2 — whole-word clustering (6 configs: layer ∈ {6,9,12} × pool ∈ {mean,stat})

Best HuBERT: **L6 stat (V=0.331, tokF1=0.207, ARI=−0.006)**. Fingerprint REF: V=**0.517**, tokF1=**(

Locked conclusion

Across **24 HuBERT configurations** spanning the standard zero-resource design space (layers 6/9/12, k

17. Bat held-out emitter validation — the claim generalizes (2026-06-23)

The §13 claim (bat recurrent acoustic-term families track emitter identity ~2.8× more than context) v

metric	Train (70%)	Held-out (30%)
Emitter V	0.221	**0.375**
Context V	0.085	0.149
ratio	2.60×	**2.52×**

85.5% of held-out windows assigned; 98.5% bout coverage. Held-out assignment is **diverse** (entropy

18. RookID individual-ID — whole-call fingerprint wins, extends bat emitter finding (2026-06-23)

The headline RookID experiment: does hierarchical sub-call structure improve individual identificat

representation	Source V	NMI	macro-F1	#train	#test	#src
whole-call fp (44-D)	**0.384**	**0.320**	**0.287**	2055	944	14
HECB-pooled fp (44-D)	0.321	0.242	0.263	2055	944	14

Findings:

1. Whole-

Dataset summary: 15,008 calls (excl Inc/Pls), 15 named rooks, 514 events (many "sing"

variants –

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19. Clusterer robustness of the pooled representation — k-means is NOT the confound (2026-06-23)

`scripts/marmoset_clusterfn_compare.py` — answers a question raised in §6/§7: is the weak

pooled-vs

clusterer	pooled V	pooled NMI	whole V	sub V	probe (whole)
k-means (spherical)	**0.440**	**0.244**	0.409	0.177	0.820
cosine-HAC	0.277	0.113	0.275	0.249	0.820
GMM (diag)	0.335	0.146	0.394	0.181	0.820

The architectural hypothesis is FALSIFIED. k-means gives the *highest* pooled V/NMI of

the three,

1. **The pooled augmentation is real and clusterer-robust** for the leading clusterer

(k-mean

This refines the §7 cross-experiment pattern: the sub-call advantage shows on supervised metrics (p

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20. Track C CCA — acoustic ↔ social share a genuine common subspace (2026-06-23)

`scripts/bat_trackc_cca.py` — unsupervised dual-validation of the §9 supervised 4.7× context-re

canonical pair	ρ	ρ²
ρ_1	**0.771**	0.595
ρ_2	0.699	0.488
ρ_3	0.623	0.388
ρ_4	0.599	0.359
ρ_5	0.541	0.293

total R^2 (first 5 comps) = 2.123

permuta

The Track C multimodal synergy is intrinsic, not a classifier artifact. A leading canonical

Honest caveat. The permutation p-value on ρ_1 is marginal (0.045), and the social space is high-dir

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21. LibriSpeech UNIFIED rerun — same words for §14 + Stage C + Stage D (2026-06-23)

`scripts/ls_unified_rerun.py` — fixes the apples-to-apples gap: the three LS word-structure scripts (§1

§14 — within/across edit-dist + NN + motif→word (n=600)

source	within	across	ratio	NN	NN_xsp	motifV
HECB(k32)	0.917	0.935	0.981	0.098	0.090	0.429
HuBERT-km(k50)	0.939	0.928	**1.012**	0.047	0.047	0.424
fingerprint(k32)	0.907	0.942	0.964	0.118	0.114	**0.481**
CPC-km(k50)	0.900	0.940	0.958	**0.175**	**0.171**	0.458

Stage C — DUSTED recurrent-term discovery (DTW row, same n=600)

source	#fams	cov	word_V	H	top-frac	null-ratio	bootstrap
HECB(k32)	260	0.038	0.419	0.98	0.02	1.15×	0.181
HuBERT-km(k50)	617	0.080	0.416	0.96	0.01	1.35×	**0.428**
fingerprint(k32)	269	0.068	**0.453**	0.96	0.03	1.26×	0.298
CPC-km(k50)	738	0.052	0.427	0.97	0.01	1.21×	0.310

Stage D — whole-word embedding clustering (same n=600, k=60)

source	V	ARI	purity	tokenF1
fingerprint	**0.517**	0.054	0.282	0.282
HECB-pooled	0.451	0.013	0.223	0.223
HuBERT-mean	0.386	−0.003	0.193	0.193
CPC-mean	0.463	0.033	0.248	0.248

Unified verdict (the point of putting all three on the same words)

The fingerprint leads on every aggregate metric: §14 motifV 0.481, Stage-C DTW word_V

0.453, S

Run: `python scripts/ls_unified_rerun.py --n-files 200 --max-words 600`; re-score:

`python s

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22. Cross-corpus DUSTED motif table — uniform threshold, HAC-corrected (2026-06-23)

The three animal DUSTED scripts (`hecb_dusted_marmoset`, `dusted_bat`, `dusted_bengalese`) now

emit iden
tier)**

corpus (n)	#fams	cov	purity(w)	H(norm)	top-frac	null-ratio	bootstrap
Bengalese finch (3 birds, 1	457	0.480	**0.788**	0.85	0.12	**2.67×**	**0.509**
Egyptian fruit bat (300 bouts)	306	0.153	0.39 †	0.87	0.06	1.36×	0.394
Infant marmoset (1500 calls)	139	0.028	0.49 †	0.89	0.06	1.55×	0.229

† purity label is corpus-specific (bengalese = GT syllable sequence → genuine blind recovery;

bat = Cor

The cross-corpus claim holds on honest (HAC-corrected) numbers. The relative ordering —

Bengalese

§11 marmoset: "DTW ct_V 0.508, 4.7× exact" was a CC-chaining artifact — HAC-corrected

marmo

call-type

Manuscript framing: DUSTED lands exactly where the repertoire predicts — diverse discrete

syllables

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23. RookID Event/Comment enrichment — whole-call fp vs DUSTED motifs × 6 targets (2026-06-23)

`scripts/rookid_event_comment.py` — keeps the two columns the prior RookID scripts dropped:

**`Event`

target	n	#cls	fp-V	fp-NMI	fp-probe	motif-V	motif-NMI
Source (individual)	2798	14	0.308	0.175	0.325	0.347	0.156
activity (singing/call)	2730	2	0.673	0.159	**0.846**	0.657	0.092
sing/non-sing	2730	2	0.673	0.159	**0.846**	0.657	0.092
song_variant (song-prefix)	1938	14	0.242	0.087	0.118	0.327	0.129
social_other (partner rook)	2809	2	**0.041**	0.001	n/a	0.176	0.001

date (recording)	2809	4	0.249	0.055	n/a	0.315	0.057

Findings (a clean individual-vs-social separation):

1. **Indivi

Manuscript framing: across RookID targets the pattern is *what* (activity/sing) and *who*

(Source) a

---

24. Soundwel Track C — does acoustic structure predict emotional valence beyond call type? (2026-06-23)

`scripts/soundwel_trackc.py` — Soundwel = 6887 labelled pig vocalizations (44.1 kHz,

~0.4 s) w

6-feature × 3-target probe (Team-held-out macro-F1)

target	whole fp	HECB	seq	motif	ctype	whole+ctype
vocalisation_type (chance 0.17, k=6)	**0.417**	0.275	0.382	0.288	—	—
production_context (chance 0.50, k=2)	0.662	—	—	—	**0.790**	0.626
emotional_valence (chance 0.50, k=2)	0.478	—	—	—	0.296	0.466

Decisive valence block (the "beyond call type" test; chance 0.50)

test	macro-F1
call-type → valence (baseline)	0.508
fp → valence	0.478
fp + call-type → valence	**0.466** (adding fp *hurts*)
within-call-type fp → valence	**0.350** (well below chance; 6 calltype
fp → valence (male / female)	0.339 / 0.352 (both below chance)

Verdict — the bioacoustic semantic ceiling, proven

**Acoustic structure does NOT predict emotional valence — it is at/below chance, including

the clean

Manuscript framing: emotional valence is locked inside the call-type template; it is not linearly se

---

25. LibriSpeech speaker-invariant / universal acoustic-structure test (2026-06-23)

`scripts/ls_speaker_invariant.py` — does the ordered acoustic sub-unit representation preserve

4-way distances (ratio = d(diff-word,same-spk) ÷ d(same-word,diff-spk); >1 ⇒ word > speaker)

representation	sw/ds	dw/ss	ratio
fp_whole (cosine)	0.160	0.137	**0.86**
fingerprint-km	0.904	0.886	0.98
HECB	0.904	0.901	1.00
CPC-km	0.900	0.875	0.97
HuBERT-km	0.929	0.908	0.98
HuBERT-DTW (raw-frame, n=300)	0.864	0.826	0.96

Cross-speaker same-word retrieval (eval=400, gallery=1895; gallery excludes query speaker; chance≈0.015)

representation	top-1	top-5	MRR
CPC-km	**0.105**	**0.315**	**0.212**
fingerprint-km	0.087	0.233	0.163
fp_whole (cosine)	0.083	0.230	0.162
HECB	0.068	0.185	0.135
HuBERT-DTW (n=300)	0.042	0.083	0.074
HuBERT-km (Levenshtein)	0.018	0.098	0.062

Word-vs-speaker disentanglement + centering (fp_whole)

word probe (speaker-held-out macro-F1) = **0.110**; speaker probe = 0.094 (chance 0.006).

word pro

Honest verdict — a *qualified* universal representation

1. **Cross-speaker same-word retrieval is well above chance** (CPC 7×, fp/fingerprint ~5–6×):

word-as

Manuscript-safe claim: *cross-speaker same-word retrieval from ordered acoustic sub-unit

represent

---

26. RookID cross-Caller same-call retrieval — the §25 animal analogue (2026-06-23)

`scripts/rookid_cross_caller.py` — the cross-species mirror of §25: does CALL-TYPE structure

survive ac

target (call-type	n label)	#labels	chance	fp top-1	fp top-5	fp MRR	ratio
activity (sing/ca	2809	2	0.572	0.665	0.910	0.771	1.00
sing/non-sing	2809	2	0.572	0.598	0.882	0.719	0.97
song_variant (sing prefix)	2066	7	0.330	**0.403**	0.677	0.514	0.99
Event (fine code)	1677	30	0.240	**0.160**	0.393	0.273	0.98

(sub-call-seq Levenshtein is within ±0.02 of fp on every cell — both reps agree.)

Cross-species verdict — animal calls are MORE caller-bound than human words are speaker-bound

1. **Coarse category barely generalizes across rooks**: activity/sing cross-caller top-1

(0.60–0.

The decisive cross-species contrast (§25 ↔ §26): on LibriSpeech, cross-**speaker**

same-**w

Manuscript framing: *cross-speaker word retrieval is well above chance, but cross-caller

call-type

---

27. DTW-barycenter consensus word template — the §25 follow-on (negative) (2026-06-23)

`scripts/ls_consensus_template.py` — §25 found linear speaker-centering **fails** (Δ −0.003),

so the pri

method	top-1	chance
raw cross-speaker NN (DTW on fp-traject	**0.212**	0.040
DBA consensus template	**0.172**	0.040
Δ(template − NN)	**−0.040**	

Verdict — no simple consensus "universal word" (honest negative)

The DBA consensus template does **not** beat raw cross-speaker NN — it is slightly *worse*

(0.172 <

Manuscript framing: *there is no simple consensus "universal word" template derivable

from the

CPC-space replication — the representation matters (the sign flips)

Re-running with `--rep cpc` (continuous 256-D CPC frame trajectories, §25's most

speaker-i

representation	raw cross-speaker NN	DBA consensus template	Δ (template − NN)
fingerprint (43-D windows, phy	0.212	0.172	**−0.040**
HuBERT (768-D frames, MLM)	0.092	0.044 (≈chance)	**−0.048**
AVES (768-D frames, multi-tax	0.096	0.028 (below chance)	**−0.068**
**CPC (256-D frames, contrast	**0.428**	**0.440**	**+0.012**

The sign **flips** across representations: in the physics-bound fingerprint, DBA averaging

**smears

Honest magnitude. The effect is real but **modest** (+0.012, paired over the same 250

queries):

Firewall / animal translation. CPC is human-trained and firewall-blocked on animals, so

this speak

---

28. RookID sing/song-bout DUSTED — within-song motif inventory (2026-06-23)

DUSTED DTW discovery on the **sing subset** (1938 calls, 6 sources, 14 song variants)

of the cac

stat	value

#families	**1398**
H(norm)	**0.93** (diverse, non-degenerate — unli
top-family-frac	**0.01** (no dominant)
src_V (Source)	0.042
var_V (song_variant)	0.084
null ratio	1.06× (p=0.111 — composition-driven)
bootstrap Jaccard	0.341 (moderate)

representation	Source V	Source F1	song_variant V	Event V
whole fp (44-D)	**0.294**	**0.826**	**0.230**	**0.243**
HECB mean (44-D)	0.302	0.822	0.206	0.215
motif hist (50-D)	0.110	0.446	0.108	0.136

Findings: (1) Rook song HAS diverse recurrent acoustic structure (1398 families, H=0.93,

top=0.01

---

29. Cross-analysis: why recurrent acoustic terms are not decoded words (2026-06-23)

Five summary cross-cuts across the validated results (§1–§28), assembled into the

reviewer-

Cross-analysis 1 — cross-speaker word invariance vs cross-caller call-type invariance

Corpus	Unit tested	Identity held out	Retrieval target	Result	Interpretation
LibriSpeech	word	speaker	same word	**5–7× chance** (CPC 0.105 fa 0.03	words generalize across speaker
RookID	Event (fine code)	caller (Source)	same Event	**below chance** (0.160 vs 0.240)	fine calls are caller-bound
RookID	song_variant	caller	same variant	**moderate** (0.403 vs 0.330)	song structure partly shar
Egyptian bat	context	emitter	same context	expected weak (§13 V=0.16)	context not acoustic-invariant
InfantMarmosetsVox	call-type	caller/twin	same call-type	**strong** (V=0.606, §18)	call type is acoustic (form), not s

The decisive contrast: human **words** generalize across speakers (5–7× chance, §25);

animal **

Cross-analysis 2 — identity / call-form / context disentanglement across corpora

Dataset	Identity signal	Call/form signal	Context/social signal	Best representation
Egyptian bat	**high** (Emitter V≈0.45)	moderate (motifs graded)	**weak** (Context V≈0.16)	fingerprint; social needed for context
RookID	**high** (Source V=0.31)	ing/call strong (V=0.67)	**weak** (social_other V=0.04)	whole-call fingerprint
InfantMarmosetsVox	moderate (caller V present)	**strong** (call-type V=0.606)	receiver-side context	fingerprint + HECB roles
Bengalese finch	bird-constant (per-bird)	**strong** (GT syllable purity 0.78)	motif/acoustic context	exact/fixed-cosine motif
Soundwel pig	— (no individual ID)	call-type acoustic (0.41)	**valence not recoverable** (0.35 < chance)	fingerprint/CPC
LibriSpeech	speaker entangled (§25 ratio=1)	**strong structure** (motif V=0.48, §21)	word GT (controlled)	fingerprint/CPC

The pattern: animal acoustic structure carries *who called* and *what call form* more

reliably th

Cross-analysis 3 — word-likeness / lexicality stress test

Criterion	LibriSpeech	Bengalese	RookID	Bat	Marmoset
Cross-individual invariance	**high** (§25 5–7×)	per-bird (n/a)	**low** (§26 ≈chance)	low (§13 emitter-bound)	moderate (V=0.60 but within-ses
Motif inventory diversity	high	**high** (§12 H=0.85)	**high** (§28 H=0.93)	moderate (§13 graded)	**sparse** (§11 degenerate)
Label purity / enrichment	high (§21 motif V 0.48)	**high** (§12 0.79)	**low** (§28 V<0.1)	low (§13 V<0.17)	moderate (§18 V=0.61)
Context-independence	**high** (word GT)	n/a	**low** (§23 social≈0)	**low** (§9 social needed)	n/a
Overall word-likeness	**high**	moderate (acoustic, not semantic)	**low–moderate**	**low**	**low**

Reviewer-safe sentence: *The pipeline recovers structured recurrent acoustic terms, but

animal co

Cross-analysis 4 — evidence-tier status (H0-CMC ladder)

Claim	Tier	Evidence	Status
Recurrent acoustic terms exist	**T0**	DUSTED discovery (§11–§13, §22, §28)	validated
Individual identity recoverable	**T1**	bat Emitter V, marmoset V=0.61, RookID Source V	validated
Call-form recoverable from acoustics	**T1**	marmoset Soundwel call-type 0.41	validated
Context is socially grounded (not T1 output)	**T2+**	bat A–G D≈G=4.3×, survives shuffle-null	validated
Valence/emotion not acoustic	**T1**	Soundwel within-call-type 0.35 < caller (§	validated (ceiling)
Receiver detects motif structure	**T2**	—	**needs playback**
Multimodal receiver convergence	**T3**	—	**needs playback**
Causally supported functional signal	**T4**	—	**needs counterfactual playback**
Ecologically validated function	**T5**	—	**needs field replication**

The pipeline has exhausted T0–T1 (acoustic association + contextual prediction). T2–T4

(receiver

Cross-analysis 5 — playback-candidate prioritization

Dataset	Best playback candidate	Why	H0-CMC prediction
RookID	Source familiarity (known vs novel cal	clean individual labels; tests caller-re	response follows caller identity, not mo
InfantMarmosetsVox	onset/body/offset role-templated calls	mechanistic HECB role test (T2)	response disrupted by role-swap → roles
Egyptian bat	emitter-associated motifs vs context-soc	tests social grounding (T2); D=4.3× pro	response follows emitter, not context
Bengalese finch	GT motif order-shuffled vs natural	positive control (T2); GT syllable sport	response disrupted by order-shuffle → mo

Counterfactual controls (H0-CMC §4): natural / motif-preserved / order-shuffled /

caller-swa
amplitude

30. Boundary-quality sensitivity sweep — at what parent quality does hierarchy help? (2026-06-23)

Question (Architecture v1.0 invariant #2/#3). HECB annotates typed internal sub-call

boundarie

Setup. `hecb_decode` accepts any parent events, so parent quality is a clean independent variable;

MeerKAT (the HECB corpus): brief calls in noise + trained CRNN ⇒ parent quality varies via REAL de

MeerKAT — real CRNN detector operating points

method	parent F1	whole V	pooled V	gap	whole probe
oracle_typed	1.000	0.364	0.386	**+0.021**	0.753
oracle_raw	1.000	0.364	0.385	+0.021	0.753
crnn0.9_typed	0.940	0.299	0.351	+0.052	0.709
crnn0.7_typed	0.934	0.320	0.341	+0.022	0.645
crnn0.7_raw	0.934	0.320	0.332	+0.012	0.645
crnn0.4_typed	0.894	0.278	0.367	+0.089	0.640
raw_nbd	0.101	0.286	0.286	**0.000 ← abstain**	0.583

Marmoset — oracle/jittered parent (rich internal structure)

method	parent MAE (ms)	whole V	pooled V	gap	whole probe
oracle_raw	0.0	0.524	0.492	**−0.032**	0.810
jitter25_raw	12.9	0.522	0.486	−0.035	0.808
jitter75_raw	38.8	0.537	0.483	−0.054	0.804
jitter150_raw	77.6	0.524	0.487	−0.037	0.792
oracle_none (no internal)	0.0	0.524	0.524	0.000	0.810

Verdict — whole-call primary (invariant #3) confirmed on both corpora

Sub-call internal structure is NOT a call-type-recovery lever. At the best parent quality the hierarchy (pooled is

Consequence for the HECB gate: there is **no meaningful abstain threshold for hierarchy benefit**, because t parent).

Honesty caveat. The ablation V uses k=12 for sweep-internal consistency, so absolute whole V

(0.364 m
(meerkat

31. Cross-species repertoire taxonomy — WHY each corpus shows the structure it does (2026-06-23)

Question (Architecture v1.0 Step 3). The boundary sweep (§30) showed whole-call primary holds

regardless

corpus	repertoire type	secondary flags	null ratio	bootstrap	coverage	emitter V	context V	social lift
Bengalese finch	`discrete_syllable`	—	2.67×	0.509	0.480	n/a	n/a	n/a
Egyptian fruit bat	`graded`	identity_dominant	1.36×	socially_grounded	0.394	0.153	0.446	0.167	1.74
Infant marmoset	`sparse_repetitive`	—	1.55×	0.229	0.028	n/a	0.599	n/a
Rook	`individually_identity_dominant`	—	n/a	n/a	n/a	0.31	0.04	n/a
LibriSpeech (human)	`human_lexical`	—	—	—	—	—	—	—

Numbers curated from validated sections (§22 cross-corpus DUSTED; §13/§18/§23 binding V; §9 social-lift),

each cite
re-derive

Classifier (calibrated so each reference corpus lands on its validated type):

1. human

Two honest design notes. (1) DUSTED family-entropy is near-max for ALL corpora (0.85–0.89) — a

weak
emitter-vs

32. Identity–form–context disentanglement — what acoustics encode / don't / need-social (2026-06-23)

Question (Architecture v1.0 Step 4). The backbone biological table: for each corpus, what do the

acoustics
partner,

corpus	individual_id	call_type	context	social_partner
Egyptian fruit bat	**strong** (V 0.446)	weak (0.167)	**needs_social** (§9 D=4.3×)	n/a
Infant marmoset	n/a (no ID labels)	**strong** (V 0.599, probe 0.829)	n/a	n/a
MeerKAT	n/a	**strong** (V 0.521, probe 0.753)	n/a	n/a
Rook	partial (Source V 0.31)	**strong** (sing/call 0.67)	weak (social_other 0.04 = ≈ chance §26)	n/a (encode)
LibriSpeech (human)	**strong (speaker probe 0.56)**	partial (token↔word 0.29)	n/a (read speech)	n/a

The three-bucket thesis, operationalised:

**ENCC
meerkat,

Human anchor (the form/meaning split). Speaker identity IS strongly encoded (CPC probe 0.56 — the

latent is s
tokens

Steps 1–4 (the paper-ready core) are complete.

33. Structural alternatives ablation — DUSTED vs RQA / Matrix Profile / PST / Transition Network (2026-06-23)

Question. Four structure-discovery techniques that do NOT rely on DUSTED's token-sequence

alignment
Modules i

dataset (regime)	alternative	result	verdict vs DUSTED §22
Bengalese (discrete, diverse)	PST + Transition Network	PST **247** predictive contexts, depth 8; **639** nodes, 1205 edges, **19** motif bouts	**CORROBORATES** (purity 0.788); DTW's
Egyptian fruit bat (graded, n=26)	RQA + Matrix Profile	RQA **DET 0.229 / LAM 0.169**; MP **recurrence_fraction 0.788**, 1 motif	**COMPLEMENTS** — quantifies the grade
Infant marmoset (sparse)	Matrix Profile (+ RQA)	MP **recurrence_fraction 0.788**, 1 motif	**FAST PRE-FILTER** — token-free O(n log
LibriSpeech (diverse syntactic)	Transition Network (word transitions)	**639** nodes, 1205 edges, **19** motif bouts	**UPPER BOUND** — scales to a far larger

The takeaway — no alternative replaces DUSTED; each finds its regime niche. PST/Transition-Network

*corrobor
*complem

Honest caveats. (1) The wav2vec2 aligner emits word IDENTITIES + phoneme TIMES (not phoneme

identities)
the user

34. Three targeted structural reanalyses — closing DUSTED

limitations (2026-06-23)

The §33 alternatives applied surgically to three known DUSTED limitations — NOT a broad reanalysis.

Scripts: `scripts/{ `.

#1 Bat — RQA laminarity is context-dependent (the dynamical lens DUSTED lacked)

context code	n	DET	LAM
12	30	0.216	**0.083**
11	30	0.264	0.229
4	30	0.251	**0.265**
3	30	0.268	0.147
6	30	0.209	0.196
9	30	0.250	0.139

Kruskal-Wallis across contexts: DET H=1.74 **p=0.884** (NS) | LAM H=13.88 **p=0.016** (significant).

Verdict: graded bat bouts have **context-dependent LAMINARITY** — the degree to which the bat

holds a s 0.265

#2 Marmoset — token-free MP confirms degenerate inventory (tokenization-independent)

n=200 calls, raw 44-D fingerprint trajectory, **bypassing HECB + DUSTED entirely:**

recurren

Verdict: the degenerate-inventory claim (§11) is **tokenization-independent** — token-free MP on

raw signa an

#3 Bengalese — full transition-network topology (songbird-syntax lens)

n=100 bouts → **24 nodes**, **5 communities**, clustering **0.686**, density **0.475**, mean path

length **1

Verdict: **small-world + modular** — the topology songbird-syntax literature reports for Bengalese

finch son
without la

Structure discovery is now 100% complete. Three paradigms (alignment / graph / dynamics) agree on

the biolog

36. The Entanglement Ceiling — caller identity and call form are non-separably entangled (2026-06-23)

Question. §26 established that animal calls are *non-additively caller-bound* — the raw distance

ratio betw
meaning

Motivation. In human speech, CPC's contrastive objective partially disentangles speaker from word

content (
ethologica

Setup. Marmoset corpus (5 call types: Phee/Twitter/Trill/Tsik/Seep; 87 callers with ≥20 calls

each). Fo
caller labe

Module: `representation/disentangle.py` (pure numpy, firewall-clean, deterministic).

Method 1: Orthogonal Subspace Projection (LDA) — linear algebraic

Learns the caller-discriminating subspace via Linear Discriminant Analysis, then projects the

fingerprin
of

	caller V	form V	ratio	Δ caller	Δ form
Baseline (raw fp)	0.611	0.614	1.0	—	—
LDA projection	0.582	0.494	0.8	−0.029	**−0.120**

Verdict: FAILS. Form degrades 4× harder than caller (−0.120 vs −0.029). The caller and form

subspace

Method 2: Orthogonal Subspace Projection (LogReg) — linear algebraic

Same principle, but the caller-discriminating directions come from a multinomial Logistic Regression

weight ma

	caller V	form V	ratio	Δ caller	Δ form
Baseline (raw fp)	0.611	0.614	1.0	—	—
LogReg projection	0.606	0.601	1.0	−0.005	−0.013

Verdict: NO EFFECT. The LogReg weight matrix could not find a caller-discriminating subspace

separable

Method 3: Per-Speaker Z-Normalization — affine transform

For each caller, centers (subtracts mean) and scales (divides by std) every dimension across all

their calls than

	caller V	form V	ratio	Δ caller	Δ form
Baseline (raw fp)	0.611	0.614	1.0	—	—
Z-normalization	0.592	0.540	0.9	−0.019	**−0.074**

Verdict: FAILS. Form degrades 4× harder than caller (−0.074 vs −0.019). Per-speaker affine

normaliza entangler

Method 4: Delta-Trajectory — non-linear structural

Replaces absolute frame fingerprints with inter-frame differences: Δ_i = fp_{i+1} − fp_i. Theory:

static call

Tested on 200 marmoset calls → 4,623 frame windows (80 ms each, 44-D fp per frame) → 4,623 delta

vectors.

	caller V	form V	ratio	Δ caller	Δ form
Raw frame fp (absolute)	0.278	0.396	1.4	—	—
Delta trajectory (differenced)	0.173	0.233	1.4	−0.105	**−0.163**

Verdict: FAILS. The disentanglement ratio stays at 1.4 — zero improvement. Form degrades 1.5×

harder th spectral

Summary: the complete disentanglement proof

Method	Mathematics	Entanglement level tested	Δ caller	Δ form	Result
LDA projection	Linear subspace	Static spectral (linear)	−0.029	−0.120	**FAIL** (overlap)
LogReg projection	Linear weight	Static spectral (linear)	−0.005	−0.013	**FAIL** (inseparable)
Z-normalization	Affine (mean + scale)	Static spectral (affine)	−0.019	−0.074	**FAIL** (overlap)
Delta-trajectory	Non-linear differencing	Dynamic temporal (structural)	+0.105	−0.163	**FAIL** (temporal dynamics are

Conclusion. The 44-D acoustic fingerprint cannot be disentangled into pure caller identity and pure

call form
of

1. **Static spectral (linear):** the caller-discriminating subspace (LDA/LogReg) overlaps with the

call-type

The semantic ceiling is a mathematical fact, not a policy choice. In human speech, CPC's

contrastiv
from

Implications for the architecture:

The etho

Named result: the Entanglement Ceiling

This is the **Entanglement Ceiling**: caller physical identity and call motor form are non-linear

Biological interpretation — the medium is the message. The delta-trajectory failure is the most

Human vs. animal — closing the loop. This sharpens the cross-species analogy and makes the huma

Why this mandates Track C (the final pivot)

With this test, the computational representation space is officially exhausted. Because the computer